

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 65

Does R2 show up as an OSPF neighbor on R1? No

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

The neighbor relationship is break down because the interface is passive. Other routers cannot learn the route rules form this router.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

serial link cost 64 + Ethernet cost 1=65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__With the default of 100 Mbps, all interfaces ≥ 100 Mbps get a cost of 1. This prevents OSPF from accurately distinguishing between Fast Ethernet, Gigabit Ethernet, 10 Gb/s, etc. A higher value gives more granular costs.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: cost of R1–R3 serial link (781, because bandwidth 128 kbps) + R3 G0/0 cost (1) = 782

192.168.23.0/30: cost of R1–R3 serial link (781) + R3–R2 serial link (64, default bandwidth on R3 side) = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_cost 781 each; path cost = $781 + 781 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__The ip ospf cost 1565 command on R1's S0/0/1 makes the direct path to R3 more expensive ($1565 + 1 = 1566$) than the path through R2 ($781 + 781 + 1 = 1563$). OSPF therefore installs the lower-cost route via R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___The router ID uniquely identifies the router in the OSPF domain. If it changes, adjacencies reset and the SPF algorithm runs again, causing instability. Setting it manually (loopback or router-id command) keeps it stable.

2. Why is the DR/BDR election process not a concern in this lab?

___In this experiment, we mainly use point-to-point links with a simpler topology, and DR/BDR is only used for broadcast networks. The LAN interfaces (G0/0) connect only to PCs, not other OSPF routers

3. Why would you want to set an OSPF interface to passive?

___Because it allows the interface to stop sending OSPF hello packets and to stop establishing adjacencies, which reduces unnecessary routing update traffic and avoids leaking routing information on LANs that do not require dynamic routing updates.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___OSPF uses zones to divide the network, splitting the entire network into smaller parts through zone 0 (the backbone zone), a layered design that improves the scalability of the network.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___router ospf 1_network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

___Since EIGRP has an AD of 90, OSPF has an AD of 110, and RIP has an AD of 120, the EIGRP route with the lowest AD is selected. ___Administrative distance is checked before metric.